ONLINE EXAMINATION, ASSESSMENT & EVALUATION SYSTEM

Semester Project Proposal

Project Title: Online Examination, Assessment & Evaluation System

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1. Introduction / Background

The landscape of digital evaluation is divided into two primary sectors: academic examination and corporate recruitment. Traditionally, universities rely on Learning Management Systems (LMS) designed for course-based quizzes, while companies use highly specialized, expensive pre-employment assessment platforms. There is a growing need for a unified, scalable system that seamlessly bridges academic rigor with corporate competency tracking, enabling diverse organizations to conduct secure, dynamic, and skill-mapped assessments.

2. Problem Statement

Current assessment platforms are highly fragmented. Academic tools lack the multi-dimensional skill evaluation needed by modern employers, whereas corporate testing platforms are prohibitively expensive and rigidly structured, making them unsuitable for standard university semester exams. Furthermore, existing systems often rely on flat percentage scoring rather than mapping candidate answers to specific competencies, leaving a gap in intelligent performance analysis.

3. Existing Problems and Research Gap

Based on current market analysis, the following limitations persist:

- **Siloed Question Banks:** Questions are locked into specific courses or tests, preventing cross-organization or cross-department reuse.
- **Basic Evaluation Metrics:** Most systems provide a simple numeric score, failing to map results to granular skills (e.g., Bloom's Taxonomy, problem-solving, analytical thinking).
- **Domain Isolation:** Platforms are either strictly educational (Moodle/Canvas) or strictly corporate (TestGorilla/Mercer Mettl).
- **Interoperability and Audit Gaps:** Limited support for API integrations with Applicant Tracking Systems (ATS) and weak, non-transparent audit logs for academic compliance.

4. Proposed Solution

The proposed solution is a scalable, multi-tenant "Online Examination, Assessment & Evaluation System" designed for both academic and professional domains. It will feature a centralized,

tagged question bank, a real-time secure exam delivery engine, and an evaluation module that generates multi-dimensional competency reports rather than simple scores.

5. Objectives

1. Develop a dynamic question bank supporting diverse formats, tagged by difficulty, skill, and cognitive level.
2. Implement a secure, real-time exam engine capable of handling concurrent users with state-recovery in case of network drops.
3. Design a hybrid evaluation system (automated for objective, manual rubric-based for subjective).
4. Generate detailed analytical reports mapping candidate performance to specific competency matrices.
5. Provide API-ready architecture for future LMS/ATS integration.

6. Scope

The system will be a web-based platform encompassing organization onboarding, user role management, assessment creation, secure execution, and automated/manual grading. It will utilize browser-level anti-cheating mechanisms. Complex AI (like facial recognition) is explicitly out of scope for this individual project; instead, heuristic proctoring (browser-lock, copy-paste disable, active-window tracking) will be implemented.

7. Target Users / Stakeholders

- **System Administrators:** Manage organizations, roles, and global settings.
- **Instructors / Recruiters:** Create question banks, design assessments, set grading rubrics, and review reports.
- **Candidates / Students:** Access dashboards, take secure exams, and view performance insights.
- **Auditors / Accreditation Bodies:** Access immutable audit logs and compliance reports.

8. Core Functional Requirements

- **Organization & Role Management:** Multi-tenant architecture isolating data between different universities/companies.
- **Dynamic Question Bank:** Create, import, and tag questions (Skill, Difficulty, Domain).
- **Exam Builder:** Define exam rules, randomization, time limits, and passing criteria.
- **Online Exam Engine:** Real-time test delivery, auto-save functionality, and session recovery.
- **Evaluation Module:** Auto-grading for MCQs/Code-snippets; rubric-based interface for essay/subjective grading.
- **Dashboard & Analytics:** Skill-gap analysis, percentile rankings, and competency radar charts.

- **Basic Proctoring:** Full-screen enforcement, tab-switch logging, and copy-paste disabling.

9. Important Non-Functional Requirements

- **Scalability:** Capable of handling hundreds of concurrent exam sessions asynchronously.
- **Reliability:** Real-time state persistence to ensure zero data loss during connectivity issues.
- **Security:** JWT-based authentication, RBAC (Role-Based Access Control), and secure data transmission.
- **Usability:** Responsive, accessible UI (WCAG basic compliance) for candidates in low-bandwidth areas.

10. Key Innovative/Differentiating Features

- **Competency-Based Output:** Moving beyond "80/100" to "Strong in Data Structures, Weak in Code Optimization".
- **Hybrid Domain Support:** Usable for a university Midterm AND a corporate HR screening test.
- **Resilient Exam State:** WebSockets/Redis-backed state management ensuring students can resume exactly where they left off after a crash.

11. CCP Justification (HEC/NCEAC Alignment)

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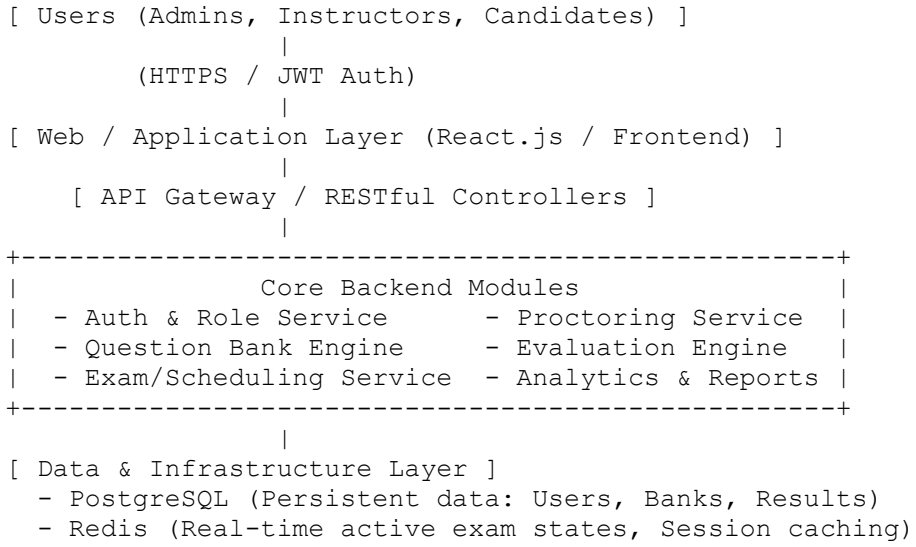
| **NCEAC CCP Characteristic** | **How the Proposed System Demonstrates It** | **System Example** | | **Conflicting Requirements** | Balancing strict anti-cheating security (which typically demands high resources/restrictions) with the need for accessibility in low-bandwidth regions. | Implementing lightweight heuristic browser-tracking instead of bandwidth-heavy live video streaming. | | **Depth of Analysis Required** | Shifting from flat arithmetic scoring to multi-dimensional skill evaluation, requiring complex data relationships. | Aggregating scores across uniquely tagged questions to generate a Bloom's Taxonomy competency radar chart. | | **Depth of Knowledge Required** | Requires advanced software engineering principles including real-time state management, secure concurrency, and multi-tenant DB design. | Using Redis for caching active exam sessions to prevent data loss and database bottlenecking during high-load exams. | | **Stakeholder Involvement** | Designing distinct, secure workflows for groups with highly varied technical literacy and objectives. | Creating separate, isolated user journeys for HR Recruiters, Academic Instructors, and Audit Officials. | | **Interdependence of Components** | The system's modules cannot function in isolation; they are tightly coupled via real-time data flow. | The Exam Engine relies instantly on the Question Bank logic, feeds data continuously to the Audit Log, and triggers the Evaluation Engine upon submission. | | **Significant Impact** | Provides a realistic, deployable solution that ensures academic integrity and fair hiring practices at a large scale. | Eliminating recruitment bias and academic fraud through standardized, auditable testing. |

12. Benchmarking with Existing Systems

| **Feature / Metric** | **Moodle / Canvas (LMS)** | **Mercer | Mettl** | **TestGorilla** | **Proposed System** | | **Primary Domain** | Academic only | Corporate & Academic | Corporate / Pre-hire | **Hybrid (Edu & Corp)** | | **Question Reusability** | Siloed by course | Proprietary / Closed | Proprietary / Closed | **Dynamic cross-org tagging** | | **Evaluation Depth** | Basic (Percentage/Grade) | Advanced | Moderate | **High (Skill/Competency Map)** | | **Proctoring / Security** | Needs paid 3rd-party plugins | High (AI-based) | High | **Baseline (Browser heuristic)** | | **Architecture / Cost** | Open-source, high setup | Expensive Enterprise | Expensive Pay-per-user | **Cost-effective, Open-API** |

Analysis: Existing enterprise tools like Mercer Mettl are too expensive for everyday university use, while LMS platforms like Moodle lack modern competency mapping and built-in assessment security. The proposed system fills this gap with an affordable, skill-focused, hybrid architecture.

13. Proposed System Architecture



14. Tools & Technologies

- **Frontend:** React.js, Tailwind CSS (for responsive, modern UI)
- **Backend:** Node.js with Express.js (for scalable, asynchronous I/O)
- **Database:** PostgreSQL (Relational integrity) and Redis (Caching/State Management)
- **Security:** JSON Web Tokens (JWT), Bcrypt (Password hashing)
- **Version Control & Deployment:** Git, GitHub, Docker (Containerization)

15. Expected Outcomes / Impact

- A fully functional, deployable assessment web application.
- Reduction in administrative overhead for instructors and recruiters.
- Enhanced fairness and transparency in grading through standardized rubrics and competency mapping.
- Fulfillment of university final-year project requirements demonstrating complex problem-solving.

16. Risks and Mitigation

- **Risk:** Real-time data loss during concurrent exam submissions.
  - **Mitigation:** Implement Redis caching for auto-saving answers every 30 seconds.
- **Risk:** Scope creep due to complex proctoring requirements.
  - **Mitigation:** Strictly limit proctoring to browser-based heuristics (tab tracking, full-screen enforcement) rather than AI video analysis.

17. Development Methodology

The project will follow an **Agile (Scrum-based)** methodology. Development will be divided into 2-week sprints, allowing for iterative testing of core modules (e.g., getting the question bank working before building the exam delivery engine) and continuous refinement based on supervisor feedback.

18. Project Timeline

- **Weeks 1-3:** Requirement Elicitation, Architecture Design, and Database Modeling.
- **Weeks 4-6:** User Authentication, Role Management, and Question Bank Module.
- **Weeks 7-9:** Exam Builder and Real-Time Assessment Engine (Core CCP focus).
- **Weeks 10-12:** Evaluation Module, Analytics, and Competency Mapping.
- **Weeks 13-14:** Security implementation (Proctoring heuristics) and System Integration.
- **Weeks 15-16:** Testing, Bug Fixing, Documentation, and Final Presentation.

19. Conclusion

The proposed Online Examination, Assessment & Evaluation System is a robust software engineering project that directly addresses the limitations of current siloed assessment tools. By bridging academic and corporate needs, implementing resilient state management, and offering deep competency analysis, the project securely satisfies the criteria for a Complex Computing Problem (CCP) and delivers a high-impact, scalable solution.